

1. Workload

2. ISA & ABI

3. Compute Org

4. Memory & NoC

5. Power Intent

6. Compiler/RT

7. Physical Node

8. Verification



CHAPTER STACK SCOPE — ACTIVE LAYERS 1, 3, 5, 7 (PHYSICAL & VERIFICATION)

Compounding Pressures & Verification Bottleneck

1. Physical Limits

- Dennard Scaling End
- Dark Silicon Thermal Wall
- SRAM Leakage Envelopes
- Sub-3nm Wire Resistance

2. Search Space

- 10^6 Candidate Combinations
- Coupled HW/SW Parameters
- Multi-Objective Trade-offs
- Non-Convex Pareto Fronts

3. Sim Bottleneck

- Signoff P&R / STA Delays
- Hours per Physical Run
- Uncalibrated Baseline Risk
- Compute Wall Bottlenecks

4. Scaling Gap

- Proposal Speed > Check Speed
- Verification Backlog Wall
- Need for AI Assistance
- Human Authority Limit

Diagnostic Telemetry & Requalification Loop